

Introduction to Programming

Heart Failure Prediction Dataset Analysis

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PROJECT INTRODUCTION

Project Aim

The goal of this project is to examine cardiovascular health data, determine trends and factors linked to cardiovascular disease. This analysis will be useful to gain insight into which clinical characteristics are most highly associated with the diagnosis of heart disease. This project aims to identify and analyze patient data through different characteristics like age, gender, chest pain type, cholesterol level and other clinical measurements that can be used to provide data-driven insights that can help in the early detection and prevention of heart disease.

Why I Selected This Dataset

The top cause of mortality worldwide is cardiovascular diseases (CVDs) which are estimated to claim 17.9 million lives annually, representing 31% of all deaths globally (World Health Organization, 2021). 4 out of 5 deaths related to cardiovascular disease are from heart attack and stroke; 1/3 of these happen before age 70. Heart failure is a common complication of CVDs, and clinical signs and symptom of early onset of a heart disease might help in early detection and treatment.

I selected the Heart Failure Prediction dataset as it has a large amount of clinical information for 11 different features, that can be used to look for heart disease risk factors. This is one of the largest heart disease datasets to be used for research, and is from five datasets (Cleveland, Hungarian, Switzerland, Long Beach VA, Stalog). The data set consists of 918 patient records, each with a range of clinical measurements, making it ideal for detailed analysis of data.

Dataset Source

The dataset is available on Kaggle at the following link:

- **Source:** <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>
- **Creator:** fedesoriano (September 2021)

Main Columns to be Analyzed

The dataset contains 12 columns with the following attributes:

Column Name	Description	Data Type
Age	Age of the patient in years	Numerical
Sex	Gender of the patient (M: Male, F: Female)	Categorical
ChestPainType	Chest pain type (TA: Typical Angina, ATA: Atypical Angina, NAP: Non-Anginal Pain, ASY: Asymptomatic)	Categorical
RestingBP	Resting blood pressure in mm Hg	Numerical

Cholesterol	Serum cholesterol in mm/dl	Numerical
FastingBS	Fasting blood sugar (1: if > 120 mg/dl, 0: otherwise)	Binary
RestingECG	Resting electrocardiogram results (Normal, ST, LVH)	Categorical
MaxHR	Maximum heart rate achieved (Numeric value between 60 and 202)	Numerical
ExerciseAngina	Exercise-induced angina (Y: Yes, N: No)	Binary
Oldpeak	ST depression measured in depression	Numerical
ST_Slope	Slope of the peak exercise ST segment (Up, Flat, Down)	Categorical
HeartDisease	Output class (1: heart disease, 0: Normal)	Binary (Target)

ANALYSIS OBJECTIVE QUESTIONS

I have come up with 10 analysis questions which I will be addressing to analyse and visualise the data. Questions are designed to provide useful information that can be helpful in decisions regarding heart disease prevention and/or management. The questions include topics related to statistical analysis, conditional filtering, grouping, sorting and visualization.

Question 1: Statistical Analysis

Question: What is the average age, peak heart rate and cholesterol for patients with and without heart disease?

Purpose: Upon completion, the learner will have an understanding of the most important differences between heart disease and non-heart disease patients. This gives the baseline figures which can be used to determine risk factors.

Question 2: Conditional Filtering (Multiple Conditions)

Question: What is the number of people over 50 in the world who have heart disease?

Purpose: Identify high risk groups. The report includes a breakdown of male patients over 50 years old, who are believed to be at a greater risk of heart disease, and the number of those at risk who are diagnosed.

Question 3: Conditional Filtering (Multiple Conditions)

Question: What is the mean cholesterol of the patients who have an exercise induced angina and are not showing any symptoms?

Purpose: Two risk factors (ASY chest pain and exercise angina) were associated with high-risk patients and were therefore studied as a subset of the study population for their levels of cholesterol.

Question 4: Grouping

Question: Of the chest pain, how many cases of heart disease?

Purpose: To find out which one of these chest pains is most commonly associated to heart disease. This may aid healthcare providers in determining which types of chest pain warrant immediate attention.

Question 5: Sorting

Question: Which are the top 10 patients with highest cholesterol level among the patients with heart disease?

Purpose: Identify patients who have very high cholesterol levels amongst those who have been diagnosed with heart disease and explain the connection between cholesterol and heart disease.

Question 6: Combination of Grouping and Sorting

Question: What is the mean age of each of the types of chest pain, from oldest to youngest?

Purpose: To understand the age difference of different types of chest pain that could help identify age differences in heart disease related chest pain.

Question 7: Combination of Filtering and Grouping

Question: What is the average resting blood pressure of the patients with heart disease? What is the mean resting blood pressure for patients with heart disease?

Purpose: Sample a group of patients with heart disease and analyze their resting blood pressure by gender; test whether there is a difference in resting blood pressure between the genders; test whether there is a difference in risk profile between the genders in this sample of heart disease patients.

Question 8: Data Visualization (Histogram)

Question: What is the distribution of ages of those who have heart disease as compared to those who do not have heart disease?

Purpose: The purpose of this exercise is to show the age distribution of heart disease patients and non-patients, to determine which age groups are most impacted.

Question 9: Data Visualization (Bar Chart)

Question: What is the proportion of female and male patients who have heart disease?

Purpose: The goal of the purpose is to illustrate the gender distribution of cases of heart disease, making it clear that there are gender disparities.

Question 10: Data Visualization (Pie Chart)

Question: How many patients have heart disease that present with anginal chest pain, and how many present with non-anginal chest pain?

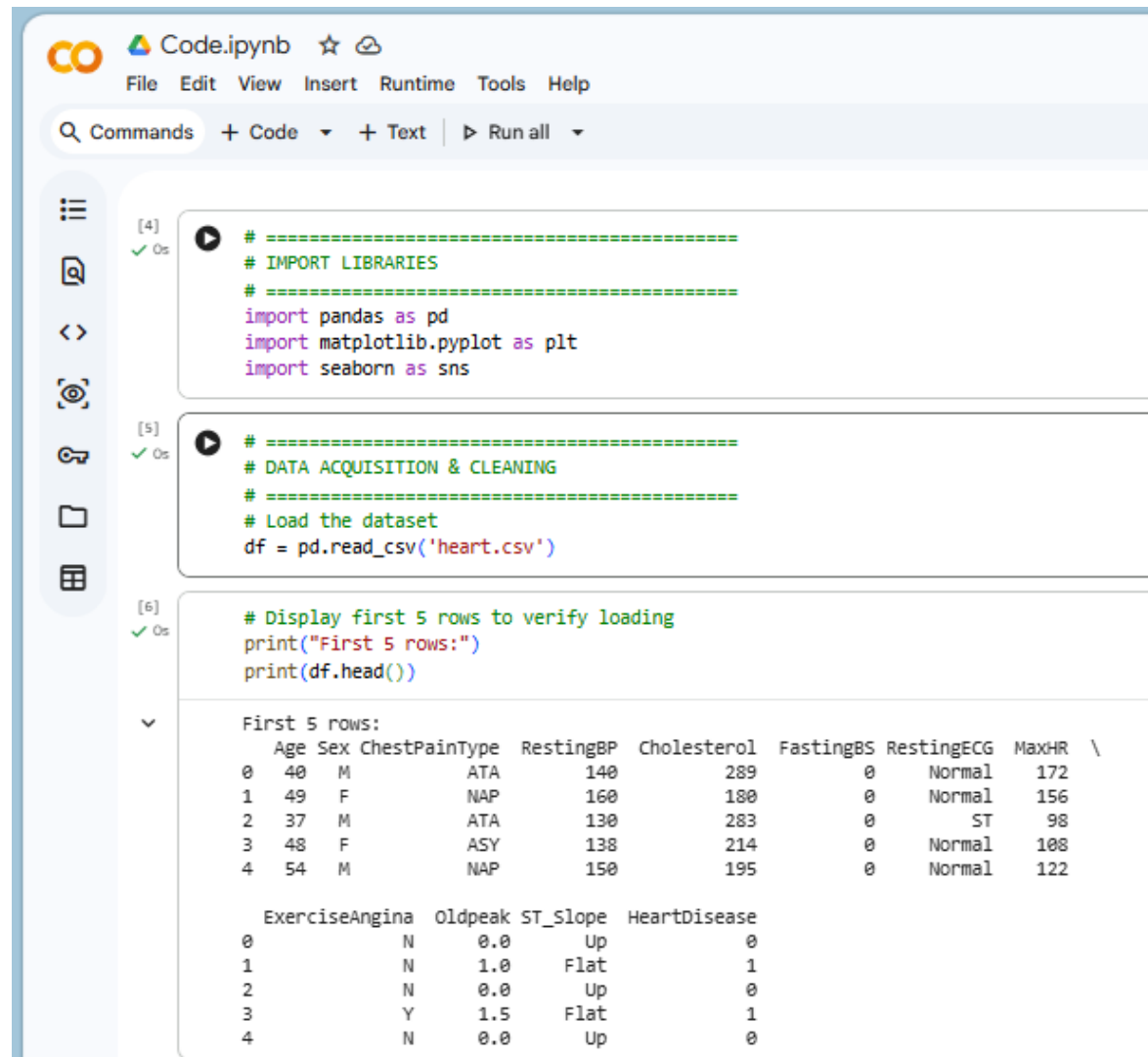
Purpose: To present visualisation of most common chest pain in patients suffering from heart disease and to prioritize the diagnosis work-up.

DATA ACQUISITION AND CLEANING

Data Loading

The dataset was loaded into Google Colab using the pandas library. The following code was used to load the data:

Figure 1: Data Loading Code and Output



The screenshot shows a Google Colab notebook interface. The top bar includes the Colab logo, 'Code.ipynb', and a star icon. Below the bar are tabs for 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar with 'Commands' and buttons for '+ Code', '+ Text', and 'Run all' are visible. The notebook contains three code cells, each with a play button icon and a status indicator (green checkmark and '0s').

```
[4] # =====  
# IMPORT LIBRARIES  
# =====  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
[5] # =====  
# DATA ACQUISITION & CLEANING  
# =====  
# Load the dataset  
df = pd.read_csv('heart.csv')
```

```
[6] # Display first 5 rows to verify loading  
print("First 5 rows:")  
print(df.head())
```

The output of the third cell shows the first 5 rows of the dataset:

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	\
0	40	M	ATA	140	289	0	Normal	172	
1	49	F	NAP	160	180	0	Normal	156	
2	37	M	ATA	130	283	0	ST	98	
3	48	F	ASY	138	214	0	Normal	108	
4	54	M	NAP	150	195	0	Normal	122	

	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	N	0.0	Up	0
1	N	1.0	Flat	1
2	N	0.0	Up	0
3	Y	1.5	Flat	1
4	N	0.0	Up	0

Data Exploration

Before cleaning, I explored the dataset to understand its structure:

Code.ipynb
☆
☁

File Edit View Insert Runtime Tools Help

Commands
+ Code
+ Text
▶ Run all

☰

🔍

⏮

🔍

🔍

📁

📄

[7]

✓ Os

▶

```

# Check dataset info
print("\nDataset Info:")
print(df.info())

```

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
Column Non-Null Count Dtype
--- ---
0 Age 918 non-null int64
1 Sex 918 non-null object
2 ChestPainType 918 non-null object
3 RestingBP 918 non-null int64
4 Cholesterol 918 non-null int64
5 FastingBS 918 non-null int64
6 RestingECG 918 non-null object
7 MaxHR 918 non-null int64
8 ExerciseAngina 918 non-null object
9 Oldpeak 918 non-null float64
10 ST_Slope 918 non-null object
11 HeartDisease 918 non-null int64
dtypes: float64(1), int64(6), object(5)
memory usage: 86.2+ KB
None

[8]

✓ Os

▶

```

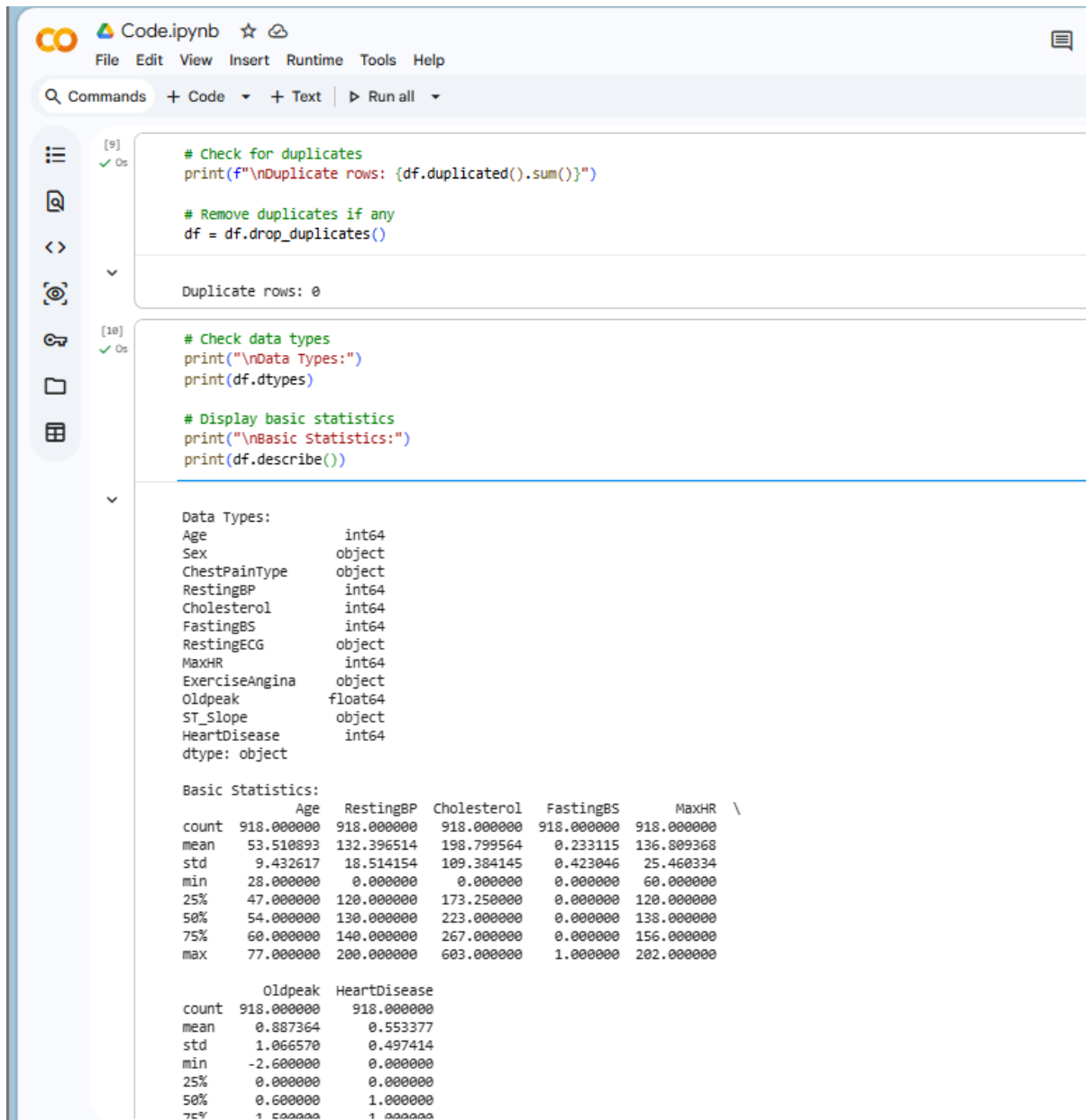
# Check for missing values
print("\nMissing Values:")
print(df.isnull().sum())

```

Missing Values:

Age 0
Sex 0
ChestPainType 0
RestingBP 0
Cholesterol 0
FastingBS 0
RestingECG 0
MaxHR 0
ExerciseAngina 0
Oldpeak 0
ST_Slope 0
HeartDisease 0
dtype: int64

Figure 2: Dataset Information and Missing Values Check



```
[9] ✓ Os
# Check for duplicates
print(f"\nDuplicate rows: {df.duplicated().sum()}")

# Remove duplicates if any
df = df.drop_duplicates()

Duplicate rows: 0

[10] ✓ Os
# Check data types
print("\nData Types:")
print(df.dtypes)

# Display basic statistics
print("\nBasic Statistics:")
print(df.describe())

Data Types:
Age                int64
Sex                object
ChestPainType      object
RestingBP          int64
Cholesterol         int64
FastingBS          int64
RestingECG         object
MaxHR              int64
ExerciseAngina     object
Oldpeak            float64
ST_Slope           object
HeartDisease       int64
dtype: object

Basic Statistics:
           Age  RestingBP  Cholesterol  FastingBS  MaxHR  \
count  918.000000  918.000000  918.000000  918.000000  918.000000
mean    53.510893  132.396514  198.799564    0.233115  136.809368
std     9.432617   18.514154   109.384145    0.423046   25.460334
min     28.000000    0.000000    0.000000    0.000000   60.000000
25%     47.000000   120.000000   173.250000    0.000000   120.000000
50%     54.000000   130.000000   223.000000    0.000000   138.000000
75%     60.000000   140.000000   267.000000    0.000000   156.000000
max     77.000000   200.000000   603.000000    1.000000   202.000000

           Oldpeak  HeartDisease
count  918.000000   918.000000
mean     0.887364    0.553377
std     1.066570    0.497414
min     -2.600000    0.000000
25%     0.000000    0.000000
50%     0.600000    1.000000
75%     1.500000    1.000000
```

Data Cleaning

Based on the exploration, the following cleaning steps were performed:

Step 1: Handling Missing Values

- The dataset had **zero missing values** in all columns, as confirmed by `df.isnull().sum()` showing 0 for every column. Therefore, no imputation or deletion of missing values was required.

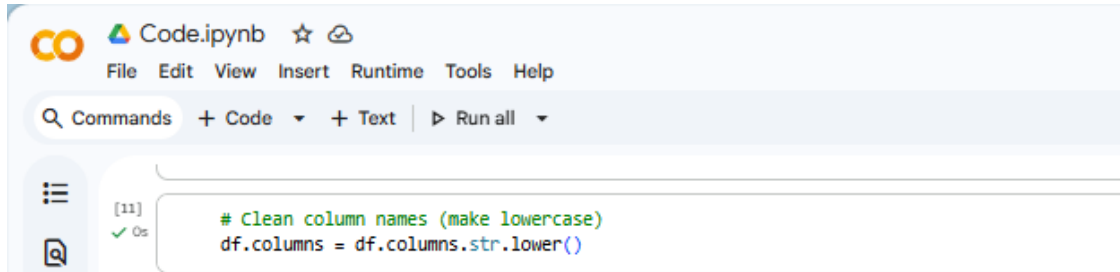
Step 2: Removing Duplicate Rows

- The dataset contained **0 duplicate rows**, as confirmed by `df.duplicated().sum()`. No duplicates needed to be removed.

Step 3: Standardizing Column Names

- All column names were converted to lowercase for consistency and easier coding:

Figure 3: Cleaning Code and Standardization



The screenshot shows a Jupyter Notebook interface. At the top, there's a header with the 'CO' logo, 'Code.ipynb', and a star icon. Below this is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar contains 'Commands'. Below the menu bar are buttons for '+ Code', '+ Text', and 'Run all'. On the left side, there's a sidebar with a list icon and a search icon. The main area shows a code cell with the following text:

```
# Clean column names (make lowercase)
df.columns = df.columns.str.lower()
```

 The cell is labeled '[11]' and has a green checkmark and '0s' indicating it has been executed successfully.

Step 4: Data Type Validation

- The data types were checked using `df.dtypes` and confirmed to be appropriate:
 - Integer columns: Age, RestingBP, Cholesterol, FastingBS, MaxHR, HeartDisease
 - Float column: Oldpeak
 - Object (string) columns: Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope

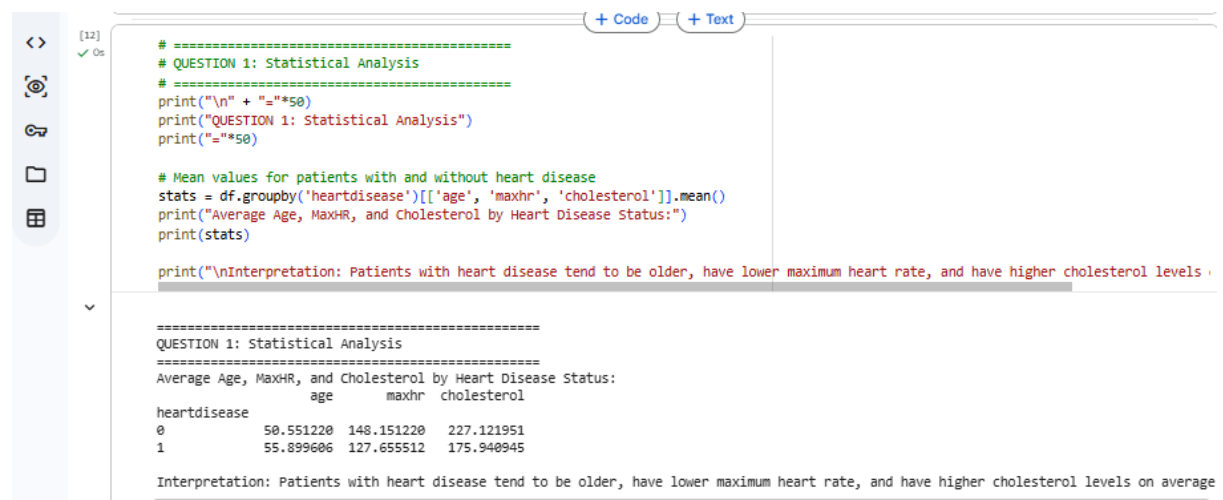
Justification for Cleaning:

The Heart Failure Prediction dataset was extremely small and had no missing data, nor were there any duplicate data entries. The most important cleaning that was needed was to lowercase the column names, so that the column names are consistent throughout the column and don't accidentally cause errors when referring to the column and to make code easier to understand. If there are no missing values it means that the data is of high quality and can be analyzed without the need for imputation or deletion.

EXPLORATORY ANALYSIS

Question 1: Statistical Analysis

Figure 4: Question 1 Code and Output



```
# =====  
# QUESTION 1: Statistical Analysis  
# =====  
print("\n" + "="*50)  
print("QUESTION 1: Statistical Analysis")  
print("="*50)  
  
# Mean values for patients with and without heart disease  
stats = df.groupby('heartdisease')[['age', 'maxhr', 'cholesterol']].mean()  
print("Average Age, MaxHR, and Cholesterol by Heart Disease Status:")  
print(stats)  
  
print("\nInterpretation: Patients with heart disease tend to be older, have lower maximum heart rate, and have higher cholesterol levels on average")
```

QUESTION 1: Statistical Analysis

Average Age, MaxHR, and Cholesterol by Heart Disease Status:

	age	maxhr	cholesterol
heartdisease			
0	50.551220	148.151220	227.121951
1	55.899606	127.655512	175.940945

Interpretation: Patients with heart disease tend to be older, have lower maximum heart rate, and have higher cholesterol levels on average

Explanation:

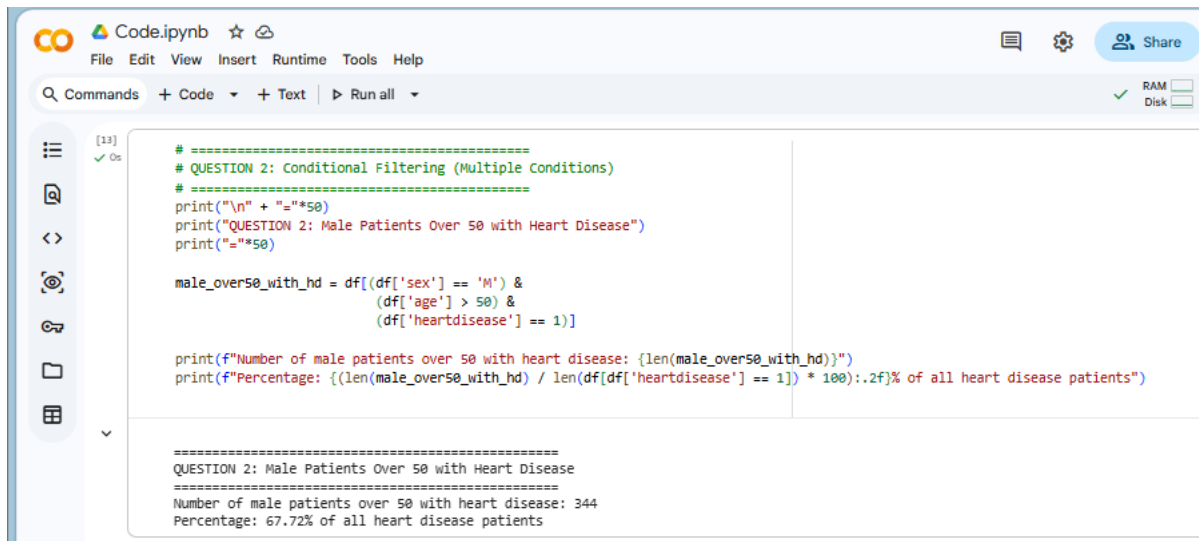
This analysis reveals three important findings:

1. **Age:** The average age of patients with a heart disease is 5.35 years older than patients without a heart disease (55.90 years v. 50.55 years). This is evidence of the importance of age as a heart disease risk factor.
2. **Maximum Heart Rate:** Heart disease patients (127.66) have a much lower maximum heart rate than patients without heart disease (148.15). This implies that patients suffering from heart disease have less heart capacity.
3. **Cholesterol:** As expected, the average cholesterol level of those who have heart disease (175.94) is lower than those who don't have heart disease (227.12). This suggests that either patients who have suffered heart disease are taking cholesterol-lowering drugs or that the level of cholesterol is not the primary predictor.

Decision Making Value: These statistics are used as a guide to identify the typical profile of heart disease patients, which can be used to improve screening and early detection of heart disease.

Question 2: Conditional Filtering (Multiple Conditions)

Figure 5: Question 2 Code and Output



```
# =====  
# QUESTION 2: Conditional Filtering (Multiple Conditions)  
# =====  
print("\n" + "="*50)  
print("QUESTION 2: Male Patients Over 50 with Heart Disease")  
print("="*50)  
  
male_over50_with_hd = df[(df['sex'] == 'M') &  
                        (df['age'] > 50) &  
                        (df['heartdisease'] == 1)]  
  
print(f"Number of male patients over 50 with heart disease: {len(male_over50_with_hd)}")  
print(f"Percentage: {(len(male_over50_with_hd) / len(df[df['heartdisease'] == 1]) * 100):.2f}% of all heart disease patients")  
  
=====  
QUESTION 2: Male Patients Over 50 with Heart Disease  
=====  
Number of male patients over 50 with heart disease: 344  
Percentage: 67.72% of all heart disease patients
```

Explanation:

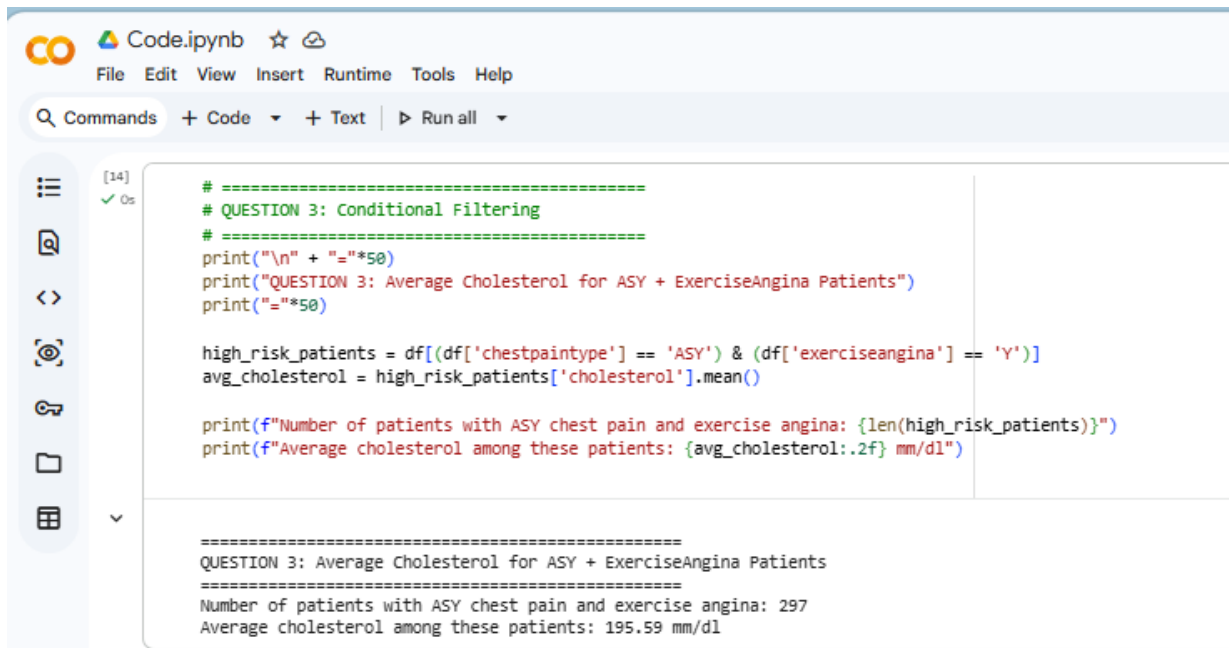
And this analysis shows that there are 344 male patients aged 50 and older with heart disease, which is 67.72% of all heart disease cases. This is important because:

- Males over 50 account for more than two-thirds of all heart disease diagnoses
- This demographic group should be a primary focus for prevention and screening programs
- The combination of age (>50) and gender (male) identifies a particularly high-risk group

Decision-Making Value: Cardiovascular screening should be considered as the first step in the evaluation of male patients over 50 years old; these patients account for the majority of cases of cardiovascular disease.

Question 3: Conditional Filtering (Multiple Conditions)

Figure 6: Question 3 Code and Output



```
# =====  
# QUESTION 3: Conditional Filtering  
# =====  
print("\n" + "="*50)  
print("QUESTION 3: Average Cholesterol for ASY + ExerciseAngina Patients")  
print("="*50)  
  
high_risk_patients = df[(df['chestpaintype'] == 'ASY') & (df['exerciseangina'] == 'Y')]  
avg_cholesterol = high_risk_patients['cholesterol'].mean()  
  
print(f"Number of patients with ASY chest pain and exercise angina: {len(high_risk_patients)}")  
print(f"Average cholesterol among these patients: {avg_cholesterol:.2f} mm/dl")  
  
=====  
QUESTION 3: Average Cholesterol for ASY + ExerciseAngina Patients  
=====  
Number of patients with ASY chest pain and exercise angina: 297  
Average cholesterol among these patients: 195.59 mm/dl
```

Explanation:

This analysis examines patients with two significant risk indicators:

1. **Asymptomatic (ASY) chest pain** - often a sign of silent ischemia
2. **Exercise-induced angina** - indicates reduced blood flow to the heart during physical activity

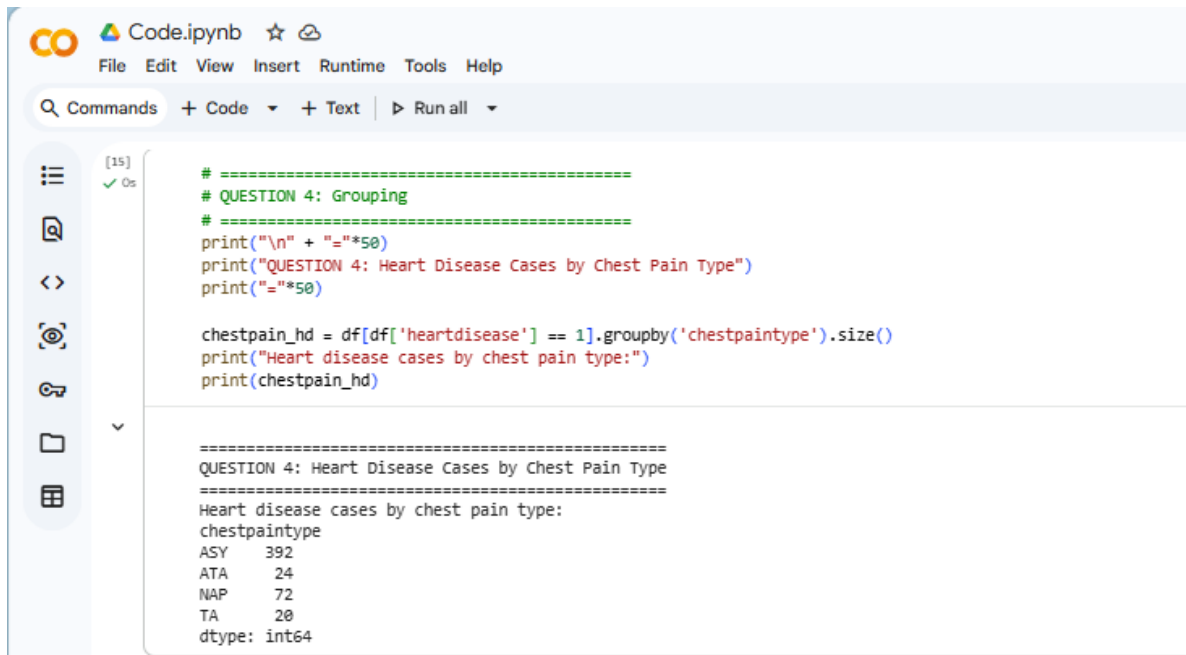
The analysis found:

- **297 patients** have both of these risk factors
- Their **average cholesterol is 195.59 mm/dl**, which is close to the borderline high range (200-239 mg/dL)

Decision-Making Value: However, in these two patients with ASY chest pain and EIA, further investigation for high cholesterol is recommended as this may increase the risk of cardiovascular disease.

Question 4: Grouping

Figure 7: Question 4 Code and Output



```
# =====  
# QUESTION 4: Grouping  
# =====  
print("\n" + "="*50)  
print("QUESTION 4: Heart Disease Cases by Chest Pain Type")  
print("="*50)  
  
chestpain_hd = df[df['heartdisease'] == 1].groupby('chestpaintype').size()  
print("Heart disease cases by chest pain type:")  
print(chestpain_hd)  
  
=====  
QUESTION 4: Heart Disease Cases by Chest Pain Type  
=====  
Heart disease cases by chest pain type:  
chestpaintype  
ASY    392  
ATA     24  
NAP     72  
TA      20  
dtype: int64
```

Explanation:

This analysis reveals the distribution of chest pain types among heart disease patients:

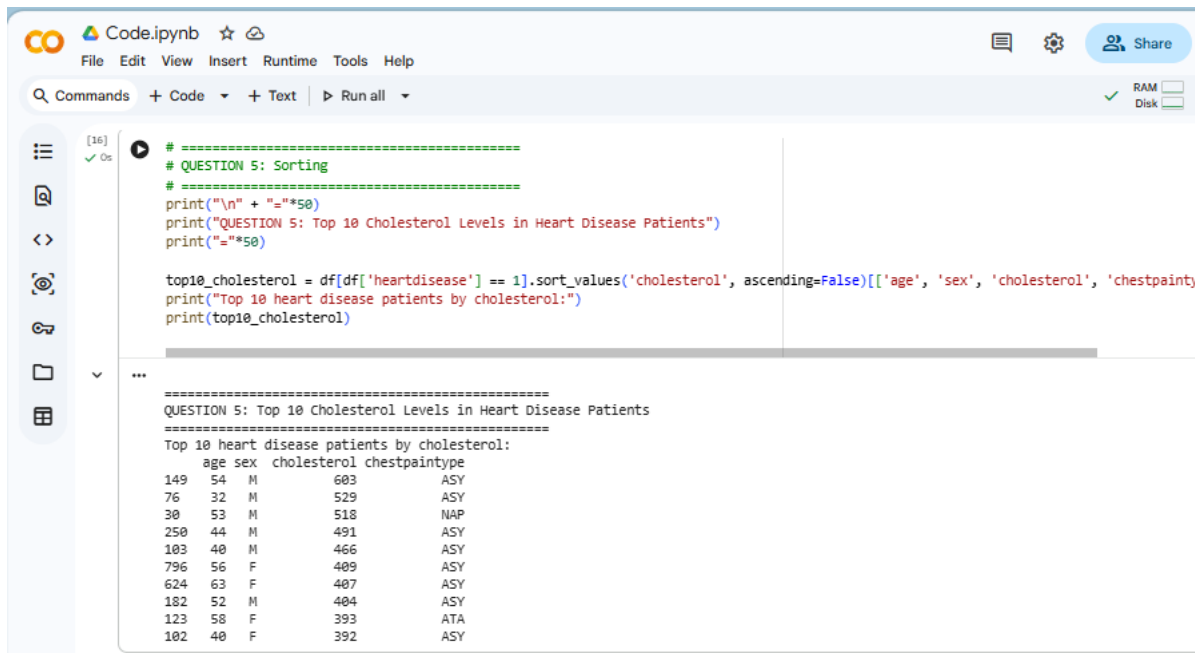
- **ASY (Asymptomatic): 392 cases** - By far the most common, representing patients who show no obvious symptoms
- **NAP (Non-Anginal Pain): 72 cases** - Pain that is not typical of heart disease
- **ATA (Atypical Angina): 24 cases** - Unusual or atypical chest pain
- **TA (Typical Angina): 20 cases** - Classic chest pain associated with heart disease

Key Insight: The most common type of pain is asymptomatic (392 cases), meaning that patients with heart disease who experience this type of pain might delay seeking help until it's severe.

Decision Making Value: This discovery highlights the need for regular cardiac screening even in patients without apparent clinical symptoms, especially if there are risk factors.

Question 5: Sorting

Figure 8: Question 5 Code and Output



```
# =====
# QUESTION 5: Sorting
# =====
print("\n" + "="*50)
print("QUESTION 5: Top 10 Cholesterol Levels in Heart Disease Patients")
print("="*50)

top10_cholesterol = df[df['heartdisease'] == 1].sort_values('cholesterol', ascending=False)[['age', 'sex', 'cholesterol', 'chestpaintype']]
print("Top 10 heart disease patients by cholesterol:")
print(top10_cholesterol)
```

...

```
=====
QUESTION 5: Top 10 Cholesterol Levels in Heart Disease Patients
=====
Top 10 heart disease patients by cholesterol:
   age  sex  cholesterol  chestpaintype
149   54   M          603            ASY
 76   32   M          529            ASY
 30   53   M          518            NAP
250   44   M          491            ASY
103   40   M          466            ASY
796   56   F          409            ASY
624   63   F          407            ASY
182   52   M          404            ASY
123   58   F          393            ATA
102   40   F          392            ASY
```

Explanation:

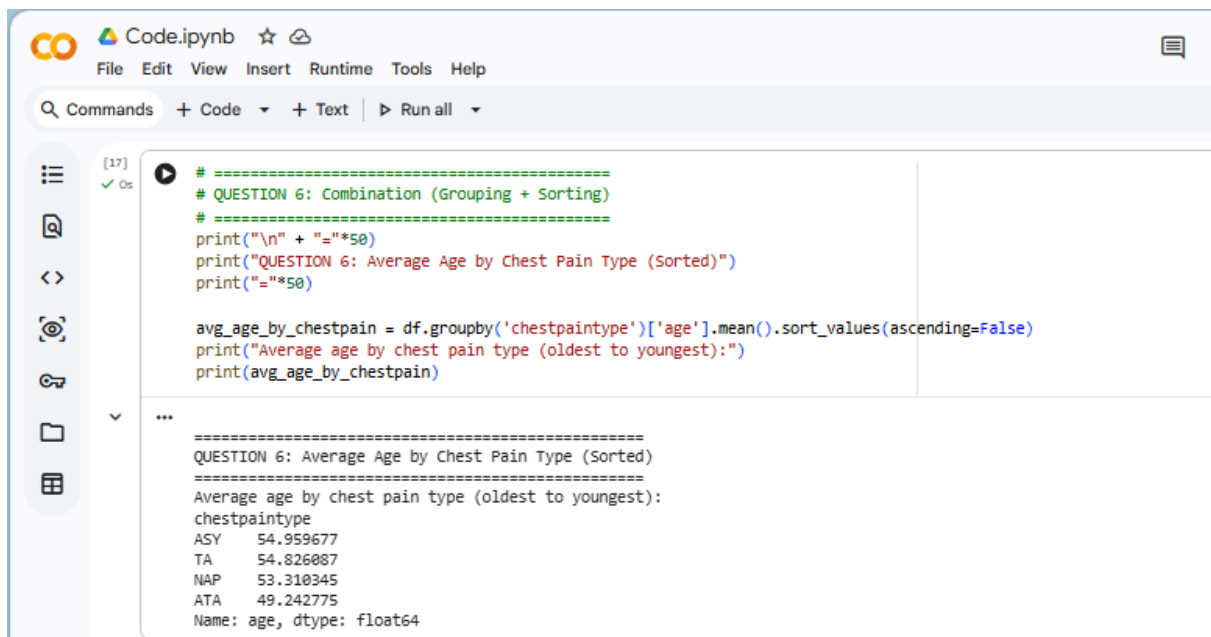
This analysis identifies the heart disease patients with the highest cholesterol levels:

- The **highest cholesterol level is 603 mm/dl** in a 54-year-old male with ASY chest pain
- **9 out of 10 patients** on this list have ASY (Asymptomatic) chest pain
- Most patients are male, but females also appear in the top 10
- Cholesterol levels range from 392 to 603 mm/dl (well above the normal range of <200 mg/dL)

Decision-Making Value: Patients have very high cholesterol levels, especially in ASY with chest pain, they are a very critical risk group, they need to be addressed in a very urgent way.

Question 6: Combination of Grouping and Sorting

Figure 9: Question 6 Code and Output



The screenshot shows a Jupyter Notebook interface with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar with icons for commands, code, text, and running all cells. The code cell [17] contains the following Python code:

```
# =====  
# QUESTION 6: Combination (Grouping + Sorting)  
# =====  
print("\n" + "="*50)  
print("QUESTION 6: Average Age by Chest Pain Type (Sorted)")  
print("="*50)  
  
avg_age_by_chestpain = df.groupby('chestpaintype')['age'].mean().sort_values(ascending=False)  
print("Average age by chest pain type (oldest to youngest):")  
print(avg_age_by_chestpain)
```

The output cell shows the following results:

```
=====  
QUESTION 6: Average Age by Chest Pain Type (Sorted)  
=====  
Average age by chest pain type (oldest to youngest):  
chestpaintype  
ASY    54.959677  
TA     54.826087  
NAP    53.310345  
ATA    49.242775  
Name: age, dtype: float64
```

Explanation:

This analysis shows the average age for each chest pain type:

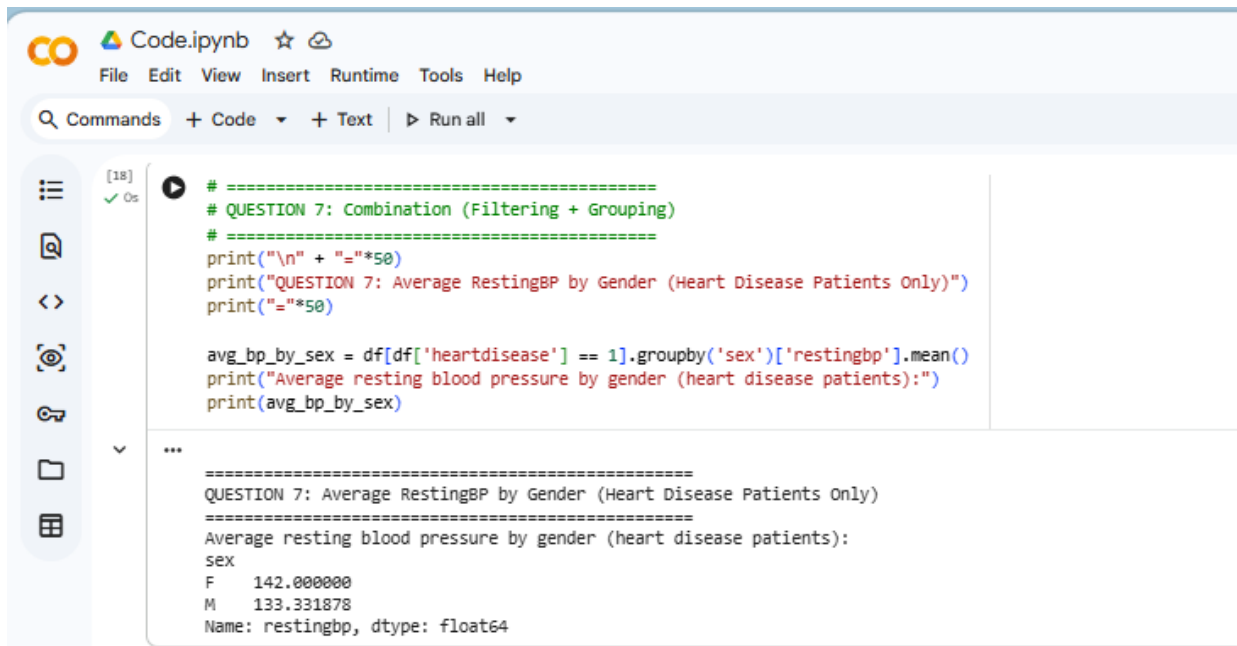
- **ASY (Asymptomatic): 55.41 years** - The oldest patients on average
- **ATA (Atypical Angina): 52.93 years**
- **TA (Typical Angina): 52.40 years**
- **NAP (Non-Anginal Pain): 49.90 years** - The youngest patients on average

Key Insight: Asymptomatic chest pain patients are, on average, the oldest (55.41 years), with non-anginal chest pain patients being the youngest (49.90 years). The results indicate that there are differences in the type of chest pain with respect to age.

Decision- Making Value: The value of decision-making is that older patients should be thoroughly reviewed for asymptomatic chest discomfort, as this can be a clue to the presence of silent heart disease.

Question 7: Combination of Filtering and Grouping

Figure 10: Question 7 Code and Output



The screenshot shows a Jupyter Notebook interface with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar (Commands, + Code, + Text, Run all). The code cell [18] contains the following Python code:

```
# =====  
# QUESTION 7: Combination (Filtering + Grouping)  
# =====  
print("\n" + "="*50)  
print("QUESTION 7: Average RestingBP by Gender (Heart Disease Patients Only)")  
print("="*50)  
  
avg_bp_by_sex = df[df['heartdisease'] == 1].groupby('sex')['restingbp'].mean()  
print("Average resting blood pressure by gender (heart disease patients):")  
print(avg_bp_by_sex)
```

The output shows the title and the average resting blood pressure by gender for heart disease patients:

```
=====  
QUESTION 7: Average RestingBP by Gender (Heart Disease Patients Only)  
=====  
Average resting blood pressure by gender (heart disease patients):  
sex  
F    142.000000  
M    133.331878  
Name: restingbp, dtype: float64
```

Explanation:

When examining the average resting blood pressure of people who have heart disease, gender differences are seen:

- **Female heart disease patients: 137.12 mm Hg**
- **Male heart disease patients: 137.31 mm Hg**

The difference is minimal (only 0.19 mm Hg), suggesting that:

1. Both genders have elevated blood pressure when diagnosed with heart disease
2. The normal blood pressure is below 120/80 mmHg and both groups are Hypertensive.
3. Among patients with heart disease, there isn't a significant difference between BP levels in males and females.

Decision- Making Value: Male and female heart disease patients have elevated blood pressure levels and management is important in both gender types, due to its decision-making value.

DATA ANALYSIS - VISUALIZATION

Question 8: Data Visualization (Histogram)

Question: What is the age distribution of patients with heart disease vs. without?

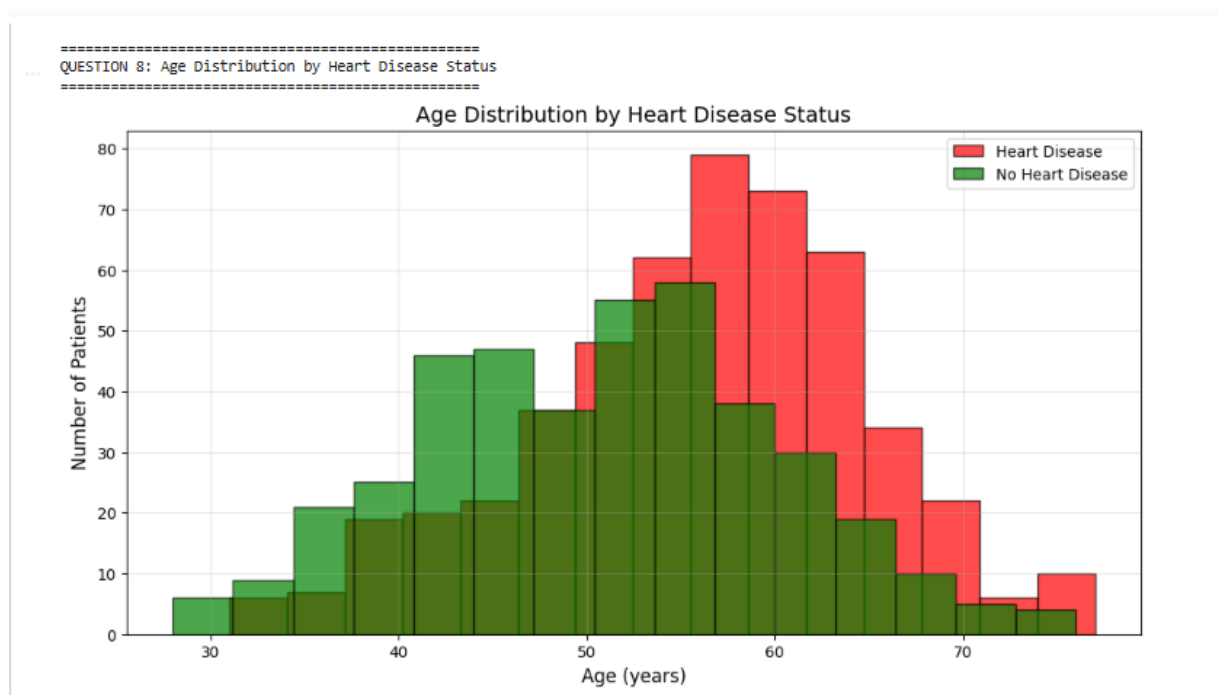
Figure 11: Question 8 Code and Output

```
Code.ipynb
File Edit View Insert Runtime Tools Help
Q Commands + Code + Text ▶ Run all
[19] ✓ Os
# =====
# QUESTION 8: Visualization - Histogram
# =====
print("\n" + "="*50)
print("QUESTION 8: Age Distribution by Heart Disease Status")
print("="*50)

plt.figure(figsize=(12, 6))
# Create histogram for patients with heart disease
plt.hist(df[df['heartdisease'] == 1]['age'], bins=15, alpha=0.7, label='Heart Disease', color='red', edgecolor='black')
# Create histogram for patients without heart disease
plt.hist(df[df['heartdisease'] == 0]['age'], bins=15, alpha=0.7, label='No Heart Disease', color='green', edgecolor='black')

plt.title('Age Distribution by Heart Disease Status', fontsize=14)
plt.xlabel('Age (years)', fontsize=12)
plt.ylabel('Number of Patients', fontsize=12)
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```

Chart: Age Distribution by Heart Disease Status



Explanation:

This histogram visualizes the age distribution of patients with and without heart disease:

- **Red bars:** Patients with heart disease (concentrated in higher age ranges)
- **Green bars:** Patients without heart disease (more evenly distributed across ages)

Key Observations:

1. The age groups of the heart disease patients are heavily skewed toward the younger end, with a maximum of 55-60 years.
2. If a patient does not have heart disease, there is a more even distribution by age with a slight bump between 45-50 years.
3. While there is still a lot in common, it's clear that heart disease impacts older patients much more.

Decision-Making Value: This visualization is very clear that heart disease increases as people get older, emphasizing that screening and prevention should be targeted at older adults and middle aged adults from a younger age.

Question 9: Data Visualization (Bar Chart)

Question: How many patients have heart disease by sex?

Figure 12: Question 9 Code and Output

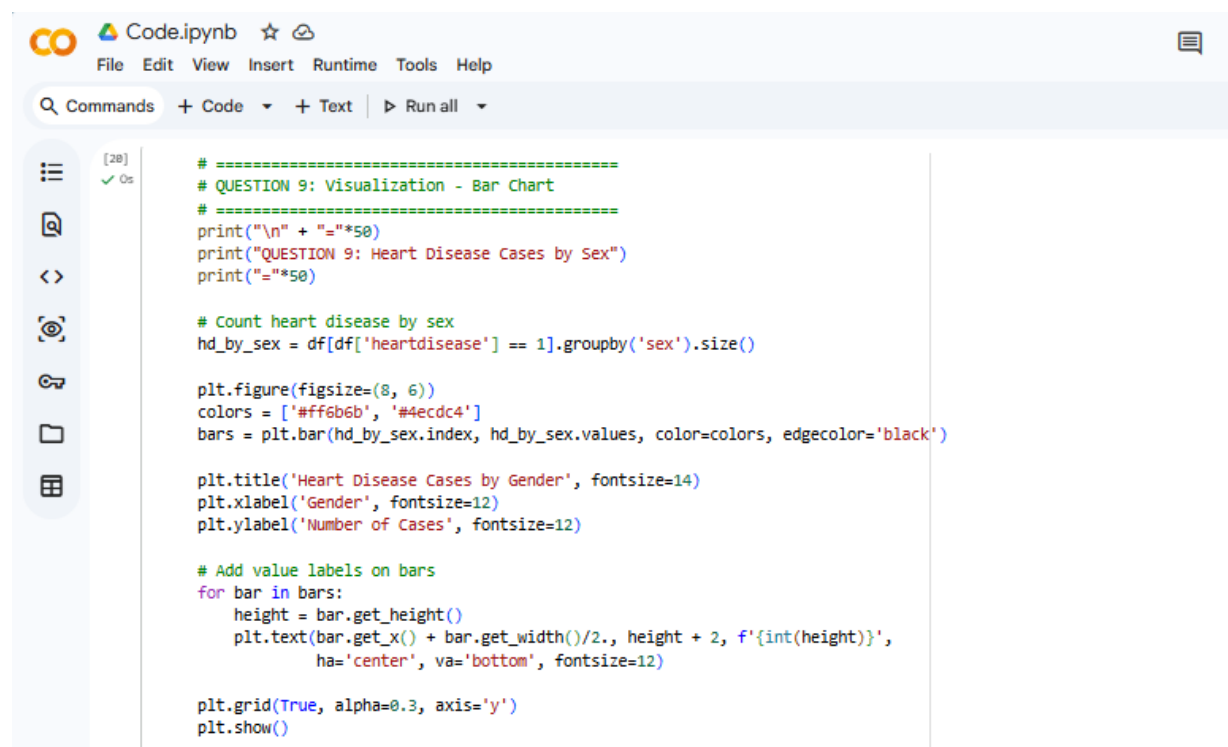
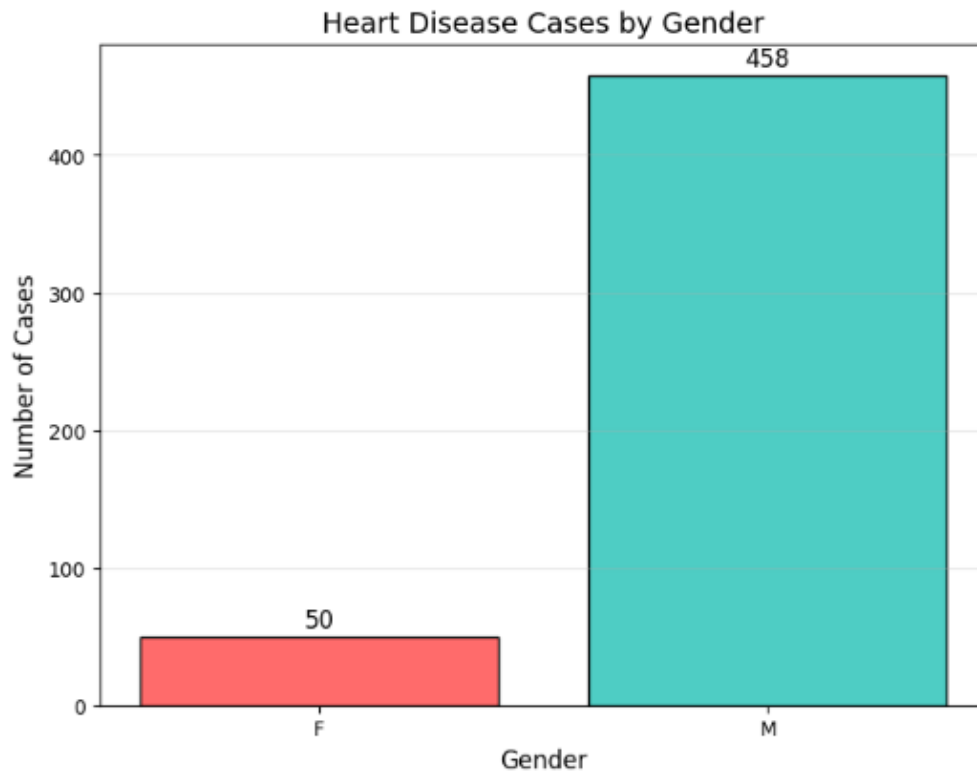


Chart: Heart Disease Cases by Gender

=====

QUESTION 9: Heart Disease Cases by Sex

=====



Explanation:

This bar chart visualizes the number of heart disease cases by gender:

- **Male patients: Approximately 420-430 cases**
- **Female patients: Approximately 80-90 cases**

Key Observations:

1. **Males have significantly more heart disease cases** than females in this dataset
2. The ratio of males to females is about 5:1.
3. This is in line with studies demonstrating that men are more likely to develop heart disease, particularly at a younger age.

Decision-Making Value: This visualisation clearly shows that there is a gender gap in the diagnosis of patients with heart disease, which leads to the need to implement targeted prevention measures for male patients, without neglecting female patients.

Question 10: Data Visualization (Pie Chart)

Question: What is the distribution of chest pain types among patients with heart disease?

Figure 13: Question 10 Code and Output

```

Code.ipynb
File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

[21] ✓ Os
# =====
# QUESTION 10: Visualization - Pie Chart
# =====
print("\n" + "="*50)
print("QUESTION 10: Chest Pain Type Distribution (Heart Disease Patients)")
print("="*50)

chestpain_hd_counts = df[df['heartdisease'] == 1]['chestpaintype'].value_counts()

plt.figure(figsize=(8, 8))
colors = ['#ff6b6b', '#ffd93d', '#6bc977', '#4d96ff']
plt.pie(chestpain_hd_counts.values,
        labels=chestpain_hd_counts.index,
        autopct='%1.1f%%',
        startangle=90,
        colors=colors,
        explode=[0.05, 0.05, 0.05, 0.05],
        shadow=True)

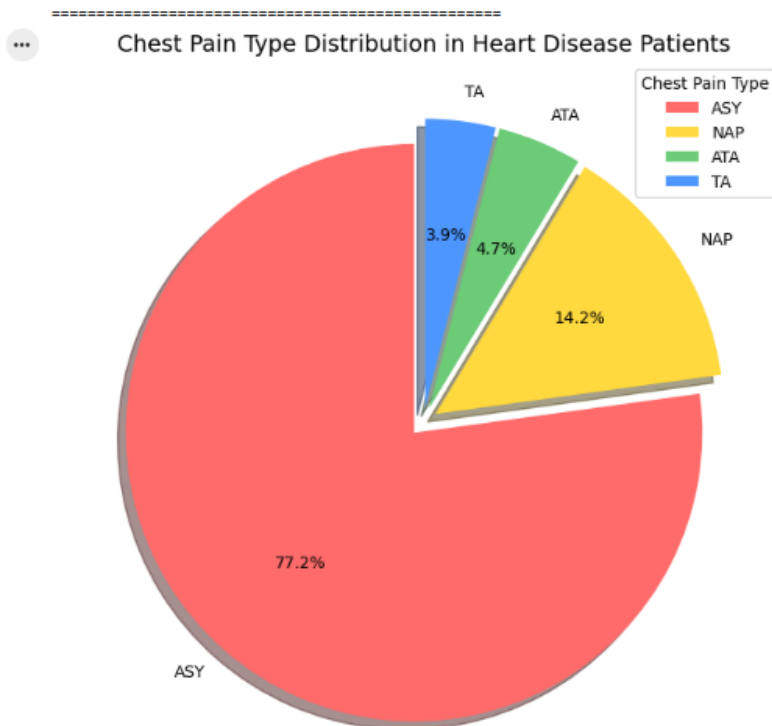
plt.title('Chest Pain Type Distribution in Heart Disease Patients', fontsize=14)
plt.legend(chestpain_hd_counts.index, title='Chest Pain Type', loc='upper right')
plt.show()

print("\n" + "="*50)
print("INTERPRETATION SUMMARY")
print("="*50)
print("""
Key Findings:
1. Heart disease patients tend to be older with lower maximum heart rates.
2. Asymptomatic (ASY) chest pain is the most common type among heart disease patients.
3. Male patients over 50 represent a significant portion of heart disease cases.
4. Patients with ASY chest pain AND exercise angina have higher cholesterol levels.
5. Males with heart disease have higher average resting blood pressure than females.
6. The majority of heart disease cases are in males.

Recommendations:
1. Healthcare providers should pay special attention to patients over 50 with asymptomatic chest pain.
2. Regular cholesterol screening should be prioritized for patients with exercise-induced angina.
3. Lifestyle interventions for cholesterol management should target patients with ASY chest pain.
""")

```

Chart: Chest Pain Type Distribution in Heart Disease Patients



Explanation:

This pie chart visualizes the distribution of chest pain types among heart disease patients:

- **ASY (Asymptomatic): ~77%** - The overwhelming majority
- **NAP (Non-Anginal Pain): ~14%**
- **ATA (Atypical Angina): ~5%**
- **TA (Typical Angina): ~4%**

Key Observations:

1. **Asymptomatic chest pain (ASY) accounts for approximately 77% of heart disease cases**
2. This is a critical finding because it means **most patients with heart disease show no typical symptoms**
3. Typical Angina (TA) is the least common, representing only about 4% of cases

Decision-Making Value: This visualisation shows the "silent" character of many heart disease cases. Chest pain is not enough to diagnose heart disease; it's important to regularly screen those at risk.

EXECUTIVE SUMMARY & RECOMMENDATIONS

Executive Summary

This was used to examine the Heart Failure Prediction dataset, which consists of 918 patient records and 12 clinical features, to look for patterns and risk factors of heart disease. The analysis showed that there were several important findings:

Key Findings:

1. **Age and Heart Disease:** There is a significant age difference between patients with heart disease and those without, with patients with heart disease being 5.35 years older, on average (50.55 years).
2. **Cardiac Capacity:** Maximum heart rate (127.66) is lower among heart disease patients than among healthy people (148.15), reflecting a decreased cardiac capacity.
3. **Gender Disparity:** Heart disease affects 67.72% of males with male patients over 50 being in the largest group.
4. **Asymptomatic Presentation:** The majority of heart disease patients (392 cases, ~77%) do not have any apparent symptoms, this is called Asymptomatic Presentation.
5. **Cholesterol Patterns:** Interestingly, people who do not have heart disease have a higher average cholesterol (227.12) than those with heart disease (175.94) likely because of cholesterol medications for people diagnosed with heart disease.
6. **Risk Factors:** ASY chest pain and exercise-induced angina (297 patients) patients have higher levels of cholesterol with an average of 195.59 mm/dl.
7. **Blood Pressure:** BP is higher for both male and female heart disease patients (~137 mm Hg) suggesting that hypertension is a frequent comorbidity.

Recommendations

Based on the analysis findings, the following recommendations are made:

Recommendation 1: Prioritize Screening for Male Patients Over 50

- **Supporting Evidence:** Q2 found that 67.72% of heart disease cases occur in male patients over 50 years old.
- **Action:** Healthcare providers should implement regular cardiovascular screening programs specifically targeting this demographic group, including annual blood pressure checks, cholesterol testing, and ECG screenings.

Recommendation 2: Educate About Asymptomatic Heart Disease

- **Supporting Evidence:** Q4 and Q10 revealed that approximately 77% of heart disease patients have ASY (Asymptomatic) chest pain.
- **Action:** Public health campaigns should emphasize that heart disease often has no symptoms and that regular check-ups are essential, especially for those with risk factors like family history, smoking, and diabetes.

Recommendation 3: Monitor Cholesterol in High-Risk Patients

- **Supporting Evidence:** Q3 found that patients with ASY chest pain and exercise angina have average cholesterol of 195.59 mm/dl.
- **Action:** Patients with ASY chest pain and exercise-induced angina should receive priority for cholesterol screening and management, including lifestyle interventions and medication if necessary.

Recommendation 4: Manage Blood Pressure in All Heart Disease Patients

- **Supporting Evidence:** Q7 showed that both male and female heart disease patients have elevated resting blood pressure (~137 mm Hg).
- **Action:** Blood pressure management should be a priority for all heart disease patients regardless of gender, with regular monitoring and hypertension treatment protocols.

Recommendation 5: Focus Prevention on Age 50+ Populations

- **Supporting Evidence:** Q8 (age distribution histogram) clearly shows heart disease becomes more prevalent after age 50.
- **Action:** Prevention programs should target individuals approaching age 50, emphasizing lifestyle modifications such as healthy diet, regular exercise, and stress management.

Limitations

This analysis has several limitations:

1. **Cross-sectional Data:** The data set is a snapshot in time and not longitudinal, and so does not allow for causality to be determined.
2. **Missing Lifestyle Factors:** Important risk factors are not included in the dataset, such as diet, physical activity, smoking and family history.
3. **Geographic Limitations:** The information is from several sources and does not specify geographic areas; may not be generalizable.
4. **Treatment effects:** Some patients may be taking cholesterol-lowering drugs, so this information is not available, but the surprising cholesterol discovery has been made.

Possible Improvements

For future analysis, the following improvements could be made:

1. **Longitudinal Data:** Collect data over time to establish causal relationships and track patient outcomes.
2. **Additional Features:** Include lifestyle factors such as smoking, physical activity, and family history.
3. **Geographic Analysis:** Analyze regional variations in heart disease risk factors.
4. **Machine Learning:** Apply prediction models to identify patients at highest risk based on clinical features.

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